

YAMAGUCHI, Japan

WMO: 477840

Lat: 34.1625N

Lon: 131.4622E

Elev: 65

StdP: 14.66

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.3	30.0	17.2	13.1	39.9	19.6	14.7	39.9	15.9	41.0	14.1	41.6	3.1	60	0.364

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.3	93.5	78.1	91.5	77.4	89.4	76.4	79.8	89.7	78.7	87.6	77.8	86.0	7.2	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.3	142.3	83.6	76.3	137.8	83.1	75.4	133.6	82.5	43.4	89.9	42.3	87.9	41.4	86.2	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.4	12.1	10.6	24.7	96.4	2.6	2.1	22.9	98.0	21.4	99.2	19.9	100.4	18.0	102.0
			23.5	81.0	2.5	1.6	21.7	82.1	20.2	83.0	18.8	83.9	17.0	85.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	61.1	40.6	42.5	48.3	57.7	66.6	73.3	80.7	82.6	75.6	65.1	54.1	44.6	(6)
(7)		DBStd	15.24	4.33	5.28	5.59	5.58	4.02	3.64	4.04	3.33	4.86	5.18	5.94	5.23	(7)
(8)		HDD50	825	292	218	100	9	0	0	0	0	0	25	181		(8)
(9)		HDD65	3178	756	629	517	225	29	0	0	0	62	328	632		(9)
(10)		CDD50	4868	2	8	48	238	515	698	950	1012	767	467	149	14	(10)
(11)		CDD65	1748	0	0	0	5	79	248	485	547	318	64	2	0	(11)
(12)		CDH74	15460	0	0	0	65	576	1463	4740	5944	2369	300	3	0	(12)
(13)		CDH80	5460	0	0	0	5	98	334	1829	2487	685	22	0	0	(13)
(14)	Wind	WSAvg	3.9	3.6	3.8	4.4	4.5	4.5	3.8	4.2	4.3	3.9	3.4	3.2	3.4	(14)
(15)	Precipitation	PrecAvg	71.8	3.1	3.3	5.2	6.1	7.4	10.5	12.0	7.4	6.7	3.5	3.6	3.0	(15)
(16)		PrecMax	99.6	6.9	5.4	9.8	12.9	15.2	17.7	27.8	20.8	15.8	9.1	9.1	5.6	(16)
(17)		PrecMin	48.4	1.1	1.0	2.3	1.9	2.5	2.1	2.1	1.7	1.7	0.4	1.1	0.4	(17)
(18)		PrecStd	13.2	1.5	1.1	2.0	2.6	3.0	4.7	7.5	4.5	3.4	2.4	2.0	1.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.0	63.9	70.2	80.1	86.7	88.9	96.3	96.8	91.4	82.7	73.6	63.2	(19)
(20)		MCWB	51.2	55.3	55.9	62.2	64.9	73.4	79.5	79.4	76.6	71.0	62.7	56.0		(20)
(21)		2%	DB	54.6	59.5	66.0	75.5	82.6	86.1	93.3	94.2	88.7	79.7	70.2	59.8	(21)
(22)		MCWB	48.0	51.5	54.1	59.2	64.1	70.9	78.0	78.5	74.8	67.9	60.9	52.4		(22)
(23)		5%	DB	52.0	55.9	62.7	72.0	79.9	83.8	91.2	92.3	86.4	77.4	67.6	57.2	(23)
(24)		MCWB	45.4	48.4	52.8	57.6	62.5	70.1	77.5	77.6	73.9	66.3	59.4	50.4		(24)
(25)		10%	DB	49.4	52.8	59.6	69.0	77.2	81.6	89.0	90.3	84.0	75.0	65.0	54.5	(25)
(26)		MCWB	43.2	46.1	51.4	57.5	62.0	69.3	76.8	77.0	72.7	64.7	57.8	48.9		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.1	58.4	60.1	67.2	71.1	77.0	81.0	81.3	78.7	74.6	66.6	58.5	(27)
(28)		MCDWB	56.4	62.4	63.3	71.9	76.5	83.5	91.9	92.4	87.3	79.3	69.4	60.2		(28)
(29)		2%	WB	49.5	53.6	57.5	63.9	69.1	75.4	79.7	80.0	76.9	71.9	63.8	54.6	(29)
(30)		MCDWB	52.4	57.3	62.3	69.5	74.2	81.1	89.7	90.2	84.2	76.7	67.9	57.6		(30)
(31)		5%	WB	46.5	50.1	55.1	61.6	67.1	74.0	78.6	79.1	75.8	69.0	61.3	51.8	(31)
(32)		MCDWB	50.2	54.2	60.7	68.1	73.5	79.1	87.9	88.0	82.9	74.4	65.7	55.5		(32)
(33)		10%	WB	44.2	47.0	52.4	59.8	65.4	72.6	77.6	78.2	74.8	66.4	58.9	49.3	(33)
(34)		MCDWB	48.6	51.8	58.3	66.0	72.8	78.0	86.0	86.5	81.5	72.2	64.1	53.9		(34)
(35)	Mean Daily Temperature Range		MDBR	14.9	16.1	18.1	19.3	18.9	14.5	13.2	14.3	14.7	17.3	17.3	15.7	(35)
(36)		5% DB	MCDBR	19.0	20.8	23.8	25.5	25.4	19.4	17.5	17.3	16.6	19.5	19.3	19.2	(36)
(37)			MCWBR	12.6	13.3	13.8	12.5	10.8	7.8	6.0	5.7	6.1	9.1	10.9	12.6	(37)
(38)			MCDBR	16.9	16.6	17.9	17.4	15.8	12.1	15.9	15.5	13.4	13.6	15.2	16.5	(38)
(39)		5% WB	MCWBR	13.0	12.9	12.9	10.4	8.8	6.0	5.8	5.6	5.6	7.8	10.1	12.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.442	0.486	0.555	0.582	0.582	0.596	0.596	0.555	0.515	0.469	0.444	0.433	(40)	
(41)		taud	2.012	1.932	1.783	1.773	1.829	1.878	1.923	2.033	2.090	2.112	2.094	2.047	(41)	
(42)		Ebn at Noon	231	235	229	232	233	230	228	236	239	239	229	225	(42)	
(43)		Edh at Noon	44	53	66	70	67	64	61	53	48	43	40	40	(43)	
(44)	All-Sky Solar Radiation	RadAvg	649	886	1226	1516	1668	1449	1539	1616	1276	1073	784	602	(44)	
(45)		RadStd	58	81	92	130	186	150	230	174	129	111	65	40	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+83
(47)	Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page